



Cinvestav
Guadalajara



COCYTEN
CONSEJO DE CIENCIA Y TECNOLOGÍA
DEL ESTADO DE NAYARIT



Nayarit
NUESTRA DIGNIDAD Y COMPROMISO



Talento Altamente Especializado – Inteligencia Artificial 2026 COCYTEN-Nayarit

Lab 5: Backpropagation Exercises: Forward and Backward Computational Graphs (**Deep Learning**)

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Introduction

Solve the backpropagation exercises: Forward and Backward Computational Graphs

Laboratory Description

Instructions

For each exercise, the model is given as

$$f(x, \theta),$$

where x is the input and θ is the set of learnable parameters.

For every exercise you must:

1. Decompose $f(x, \theta)$ into elementary forward-pass computations.
2. Compute the loss
$$\ell = (f(x, \theta) - y)^2.$$
3. Draw the forward computational graph.
4. Derive the backward-pass computations using the chain rule.
5. Draw the backward computational graph with gradients on the edges.
6. Compute the gradients with respect to all learnable parameters in θ .



Exercise 1: Linear Model

Model:

$$f(x, \theta) = wx + \beta, \quad \theta = \{w, \beta\}.$$

1.- Forward Pass $\rightarrow$ Descomponemos el modelo

$$\begin{aligned} z_1 &= wx \\ z_2 &= z_1 + \beta \\ f &= z_2 \\ e &= f - y \\ \mathcal{L} &= e^2 \end{aligned}$$

2.- Loss $\rightarrow$ Sustituyendo $f(x, \theta) = wx + \beta$

$$\mathcal{L} = (wx + \beta - y)^2$$

3.- Gráfico Computacional $\rightarrow$ "Se envía en imagen"

4.- Backward pass usando regla de la cadena

1: $\frac{\partial \mathcal{L}}{\partial e} = 2e$

Como: $e = f - y$ entonces $\frac{\partial e}{\partial f} = 1$

• (por lo tanto)

$$\frac{\partial \mathcal{L}}{\partial f} = 2e; \text{ si sustituimos } e = f - y$$

$$\frac{\partial \mathcal{L}}{\partial f} = 2(f - y) \rightarrow f = wx + \beta$$

tenemos: $\frac{\partial f}{\partial w} = x$ y $\frac{\partial f}{\partial \beta} = 1$

es decir: $\frac{\partial \mathcal{L}}{\partial y} = -2(f - y) \rightarrow \frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial w} \rightarrow \frac{\partial \mathcal{L}}{\partial \beta} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial \beta}$



5.- Grafo "backward con gradientes" → "se envió por correo"

6.- Gradientes finales:

para "w": $\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial w}$; $\frac{\partial \mathcal{L}}{\partial w} = 2(f-y)x$

"sustituyendo" → $f = wx + \beta$:

$$\frac{\partial \mathcal{L}}{\partial w} = 2(wx + \beta - y)x$$

para " β ": $\frac{\partial \mathcal{L}}{\partial \beta} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial \beta}$; $\frac{\partial \mathcal{L}}{\partial \beta} = 2(f-y)(1)$

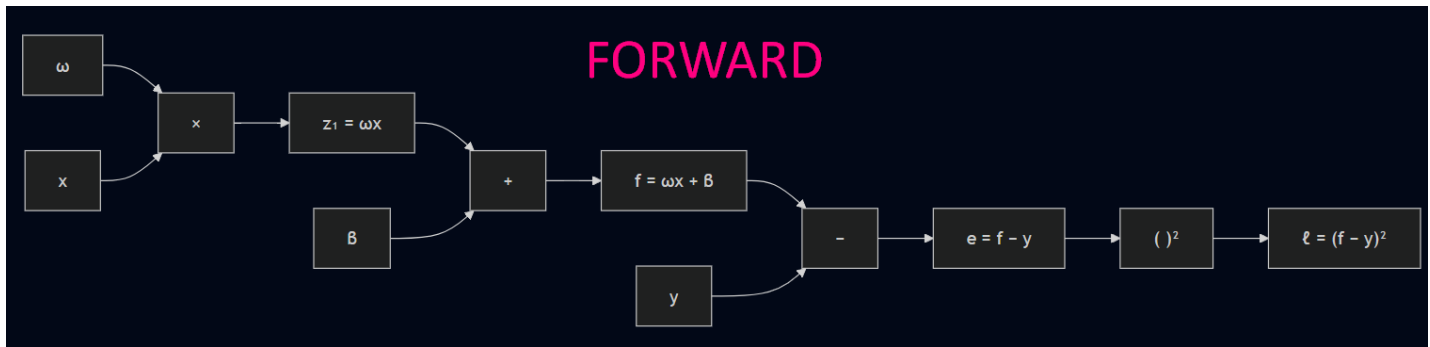
$$\frac{\partial \mathcal{L}}{\partial \beta} = 2(wx + \beta - y)$$

Resultado Final: $\nabla_{\theta} \mathcal{L} = \left\{ \frac{\partial \mathcal{L}}{\partial w}, \frac{\partial \mathcal{L}}{\partial \beta} \right\} = \{2(wx + \beta - y)x, 2(wx + \beta - y)\}$



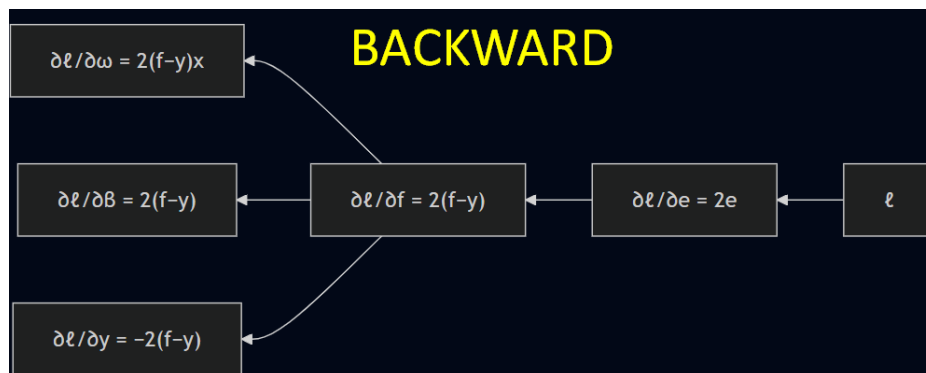
Grafo Forward

```
1 flowchart LR
2   w["ω"] --> mult["x"]
3   x["x"] --> mult
4   mult --> z1["z₁ = ωx"]
5   z1 --> sum["+"]
6   b["β"] --> sum
7   sum --> f["f = ωx + β"]
8   f --> sub[" "]
9   y["y"] --> sub
10  sub --> e["e = f - y"]
11  e --> sq["( )²"]
12  sq --> loss["ℓ = (f - y)²"]
```



Grafo Backward

```
flowchart RL
  loss["ℓ"] --> de["∂ℓ/∂e = 2e"]
  de --> df["∂ℓ/∂f = 2(f-y)"]
  df --> dw["∂ℓ/∂ω = 2(f-y)x"]
  df --> db["∂ℓ/∂β = 2(f-y)"]
  df --> dy["∂ℓ/∂y = -2(f-y)"]
```





Exercise 2: One Nonlinear Activation

Model:

$$f(x, \theta) = \tanh(\omega x + \beta), \quad \theta = \{\omega, \beta\}.$$

1.- Forward Pass :

$$\begin{aligned} z_1 &= \omega x \\ z_2 &= z_1 + \beta \\ f &= \tanh(z_2) \\ e &= f - y \\ l &= e^2 \end{aligned}$$

2.- Loss :

$$l = (f - y)^2$$

sustituyendo "f":

$$l = (\tanh(\omega x + \beta) - y)^2$$

3.- Grafo "Forward" → "Se envía como imagen"

4.- Backward Pass -

1: $\frac{\partial l}{\partial e} = 2e$ como $e = f - y$

entonces

$$\frac{\partial e}{\partial f} = 1 \quad \therefore \frac{\partial l}{\partial f} = 2(f - y)$$

$f = \tanh(z_2) \rightarrow$ la derivada: $\frac{\partial f}{\partial z_2} = 1 - \tanh^2(z_2)$

Si $f = \tanh(z_2) \rightarrow \frac{\partial f}{\partial z_2} = 1 - f^2$ entonces :

$$\frac{\partial l}{\partial z_2} = \frac{\partial l}{\partial f} \frac{\partial f}{\partial z_2} \rightarrow \frac{\partial l}{\partial z_2} = 2(f - y)(1 - f^2)$$

$$z_2 = \omega x + \beta$$

$\therefore$

$$\frac{\partial z_2}{\partial \omega} = x \quad y \quad \frac{\partial z_2}{\partial \beta} = 1$$



5.- Grafo "Backward" → se envió por correo

6.- Gradientes Finales:

para " w ":
$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial z_2} \frac{\partial z_2}{\partial w} \rightarrow \frac{\partial \mathcal{L}}{\partial w} = 2(f-y)(1-f^2)x$$

sustituyendo $f = \tanh(wx + \beta)$:

$$\frac{\partial \mathcal{L}}{\partial w} = 2(\tanh(wx + \beta) - y)(1 - \tanh^2(wx + \beta))x$$

para " β ":
$$\frac{\partial \mathcal{L}}{\partial \beta} = \frac{\partial \mathcal{L}}{\partial z_2} \frac{\partial z_2}{\partial \beta} \rightarrow \frac{\partial \mathcal{L}}{\partial \beta} = 2(f-y)(1-f^2)$$

sustituyendo $f = \tanh(wx + \beta)$:

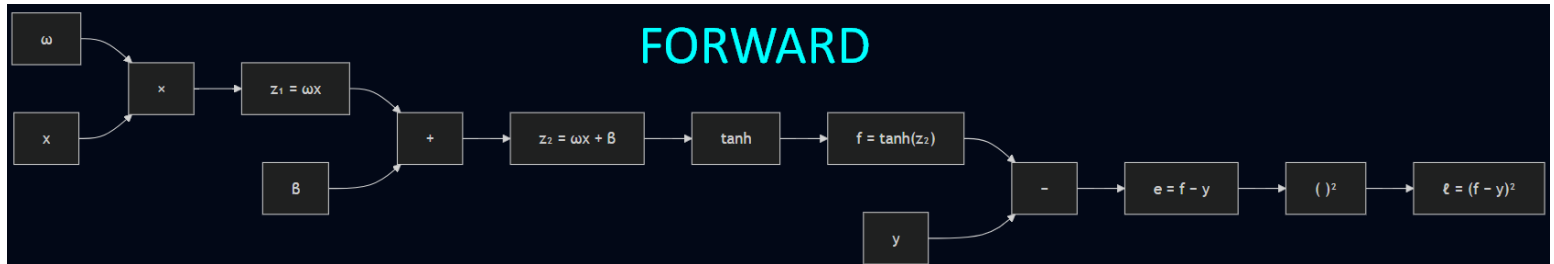
$$\frac{\partial \mathcal{L}}{\partial \beta} = 2(\tanh(wx + \beta) - y)(1 - \tanh^2(wx + \beta))$$



Grafo Forward

flowchart LR

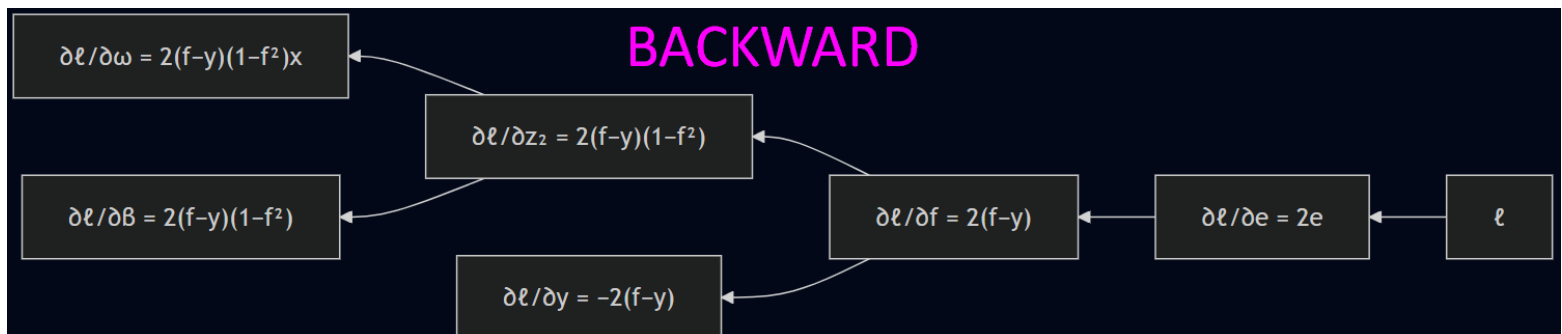
```
w["ω"] --> mult["x"]
x["x"] --> mult
mult --> z1["z1 = ωx"]
z1 --> sum["+"]
b["β"] --> sum
sum --> z2["z2 = ωx + β"]
z2 --> tanh["tanh"]
tanh --> f["f = tanh(z2)"]
f --> sub[" "]
y["y"] --> sub
sub --> e["e = f - y"]
e --> sq["( )2"]
sq --> loss["ℓ = (f - y)2"]
```



Grafo Backward

flowchart RL

```
loss["ℓ"] --> de["∂ℓ/∂e = 2e"]
de --> df["∂ℓ/∂f = 2(f - y)"]
df --> dz2["∂ℓ/∂z2 = 2(f - y)(1 - f2)"]
dz2 --> dw["∂ℓ/∂ω = 2(f - y)(1 - f2)x"]
dz2 --> db["∂ℓ/∂β = 2(f - y)(1 - f2)"]
df --> dy["∂ℓ/∂y = -2(f - y)"]
```





Exercise 3: One Hidden Neuron with Sine

Model:

$$f(x, \theta) = \omega_1 \sin(\omega_0 x + \beta_0) + \beta_1,$$

$$\theta = \{\omega_0, \beta_0, \omega_1, \beta_1\}.$$

1.- Forward Pass $\rightarrow$

$$\begin{aligned} z_1 &= \omega_0 x \\ z_2 &= z_1 + \beta_0 \\ h &= \sin(z_2) \\ z_3 &= \omega_1 h \\ f &= z_3 + \beta_1 \\ e &= f - y \\ l &= e^2 \rightarrow (f - y)^2 \end{aligned}$$

2.- Loss $l = (\omega_1 \sin(\omega_0 x + \beta_0) + \beta_1 - y)^2$

3.- Creo "Forward" $\rightarrow$ Se envia como imagen

4.- Backward Pass $\frac{\partial l}{\partial e} = 2e$ y $\frac{\partial e}{\partial f} = 1$

$$\frac{\partial l}{\partial f} = 2(f - y) \rightarrow f = z_3 + \beta_1$$

$$2(f - y) = \frac{\partial l}{\partial z_3} \text{ y tambien } \frac{\partial l}{\partial \beta_1}$$

$$\therefore$$

$$z_3 = \omega_1 h$$

$$\frac{\partial l}{\partial \omega_1} = 2(f - y)h$$

$$\frac{\partial l}{\partial h} = 2(f - y)\omega_1$$

$$\therefore$$

$$h = \sin(z_2)$$

$$\frac{\partial h}{\partial z_2} = \cos(z_2) \text{ y } \frac{\partial l}{\partial z_2} = 2(f - y)\omega_1 \cos(z_2)$$



$$\therefore z_2 = w_0 x + \beta_0 :$$

$$\frac{\partial L}{\partial w_0} = 2(f-y)w_1 \cos(z_2)x \quad y$$

$$\frac{\partial L}{\partial \beta_0} = 2(f-y)w_1 \cos(z_2)$$

5.-

Grafo "Backward" → se envía como imagen

6.- Gradientes Finales

$$\frac{\partial L}{\partial w_0} = 2(f-y)w_1 \cos(w_0 x + \beta_0)x$$

$$\frac{\partial L}{\partial \beta_0} = 2(f-y)w_1 \cos(w_0 x + \beta_0)$$

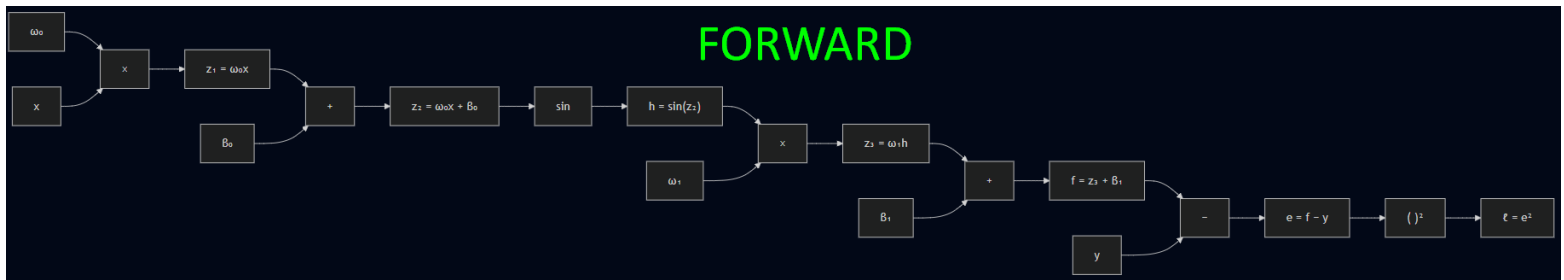
$$\frac{\partial L}{\partial w_1} = 2(f-y)\sin(w_0 x + \beta_0)$$

$$\frac{\partial L}{\partial \beta_1} = 2(f-y)$$



Grafo Forward

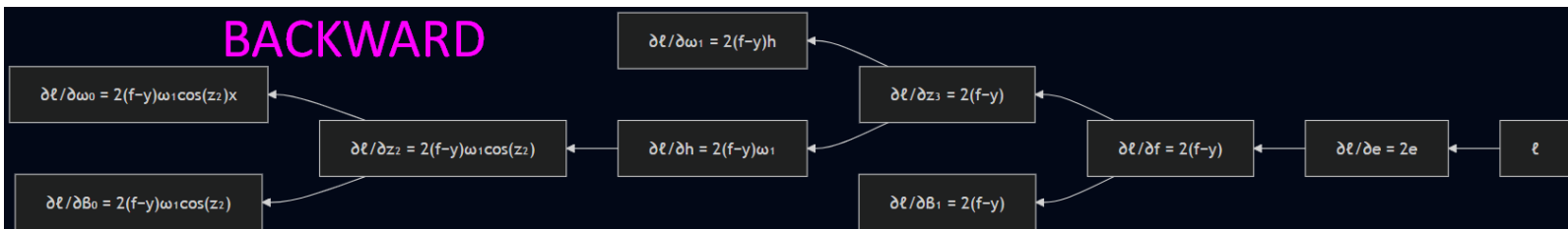
```
flowchart LR
    w0["w0"] --> m1["m1"]
    x["x"] --> m1
    m1 --> z1["z1 = w0x"]
    z1 --> s1["+"]
    b0["b0"] --> s1
    s1 --> z2["z2 = w0x + b0"]
    z2 --> sin1["sin"]
    sin1 --> h["h = sin(z2)"]
    h --> m2["m2"]
    w1["w1"] --> m2
    m2 --> z3["z3 = w1h"]
    z3 --> s2["+"]
    b1["b1"] --> s2
    s2 --> f["f = z3 + b1"]
    f --> sub["sub"]
    y["y"] --> sub
    sub --> e["e = f - y"]
    e --> sq["sq"]
    sq --> loss["loss = e^2"]
```



Grafo Backward

```
flowchart RL
    loss["loss"] --> de["de = 2e"]
    de --> df["df = 2(f-y)"]
    df --> dz3["dz3 = 2(f-y)"]
    df --> db1["db1 = 2(f-y)"]
    dz3 --> dw1["dw1 = 2(f-y)h"]
    dz3 --> dh["dh = 2(f-y)w1"]
    dh --> dz2["dz2 = 2(f-y)w1cos(z2)"]
    dz2 --> dw0["dw0 = 2(f-y)w1cos(z2)x"]
    dz2 --> db0["db0 = 2(f-y)w1cos(z2)"]
```

BACKWARD





Exercise 4: Scalar Deep MLP

Model:

$$f(x, \theta) = w_2 \tanh(w_1 \sin(w_0 x + \beta_0) + \beta_1) + \beta_2,$$

$$\theta = \{w_0, \beta_0, w_1, \beta_1, w_2, \beta_2\}.$$

1.- Forward Pass

$$f_0 = w_0 x + \beta_0$$

$$h_1 = \sin(f_0)$$

$$f_1 = w_1 h_1 + \beta_1$$

$$h_2 = \tanh(f_1)$$

$$f_2 = w_2 h_2 + \beta_2$$

$$l = (f_2 - y)^2$$

2.- Loss

$$l = (w_2 \tanh(w_1 \sin(w_0 x + \beta_0) + \beta_1) + \beta_2 - y)^2$$

3.- Grafo "Forward" → Se envia como imagen

4.- Backward Pass :

$$\frac{\partial l}{\partial f_2} = 2(f_2 - y)$$

$$f_2 = w_2 h_2 + \beta_2$$

$$\frac{\partial l}{\partial w_2} = 2(f_2 - y) h_2$$

$$\frac{\partial l}{\partial \beta_2} = 2(f_2 - y)$$

$$\frac{\partial l}{\partial h_2} = 2(f_2 - y) w_2$$

como: $h_2 = \tanh(f_1)$

$$\frac{\partial h_2}{\partial f_1} = 1 - \tanh^2(f_1)$$

$$\frac{\partial h_2}{\partial f_1} = 1 - h_2^2 \text{ entonces:}$$

$$\frac{\partial l}{\partial f_1} = 2(f_2 - y) w_2 (1 - h_2^2)$$



$$f_1 = w_1 h_1 + B_1 : \begin{cases} \frac{\partial L}{\partial w_1} = 2(f_2 - y) w_2 (1 - h_1^2) h_1 \\ \frac{\partial L}{\partial B_1} = 2(f_2 - y) w_2 (1 - h_1^2) \\ \frac{\partial L}{\partial h_1} = 2(f_2 - y) w_2 (1 - h_1^2) w_1 \end{cases}$$

$$h_1 = \sin(f_0) : \begin{cases} \frac{\partial h_1}{\partial f_0} = \cos(f_0) \\ \frac{\partial L}{\partial f_0} = 2(f_2 - y) w_2 (1 - h_1^2) w_1 \cos(f_0) \end{cases}$$

$$f_0 = w_0 x + B_0 : \begin{cases} \frac{\partial L}{\partial w_0} = 2(f_2 - y) w_2 (1 - h_1^2) w_1 \cos(f_0) x \\ \frac{\partial L}{\partial B_0} = 2(f_2 - y) w_2 (1 - h_1^2) w_1 \cos(f_0) \end{cases}$$

5.- Grafo "Backward" → Se enrí como imagen

6.- Gradientes Finales:

$$\begin{aligned} \frac{\partial L}{\partial w_0} &= 2(f_2 - y) w_2 (1 - h_1^2) w_1 \cos(f_0) \\ \frac{\partial L}{\partial B_0} &= 2(f_2 - y) w_2 (1 - h_1^2) w_1 \cos(f_0) \\ \frac{\partial L}{\partial w_1} &= 2(f_2 - y) w_2 (1 - h_1^2) h_1 \\ \frac{\partial L}{\partial B_1} &= 2(f_2 - y) w_2 (1 - h_1^2) \\ \frac{\partial L}{\partial w_2} &= 2(f_2 - y) h_2 \\ \frac{\partial L}{\partial B_2} &= 2(f_2 - y) \end{aligned}$$

donde

$$\begin{aligned} f_0 &= w_0 x + B_0 \\ h_1 &= \sin(f_0) \\ f_1 &= w_1 h_1 + B_1 \\ h_2 &= \tanh(f_1) \\ f_2 &= w_2 h_2 + B_2 \end{aligned}$$



Grafo Forward

flowchart LR

```
x["x"] --> f0["f0 = ω0x + β0"]
w0["ω0"] --> f0
b0["β0"] --> f0
```

```
f0 --> h1["h1 = sin(f0)"]
```

```
h1 --> f1["f1 = ω1h1 + β1"]
w1["ω1"] --> f1
b1["β1"] --> f1
```

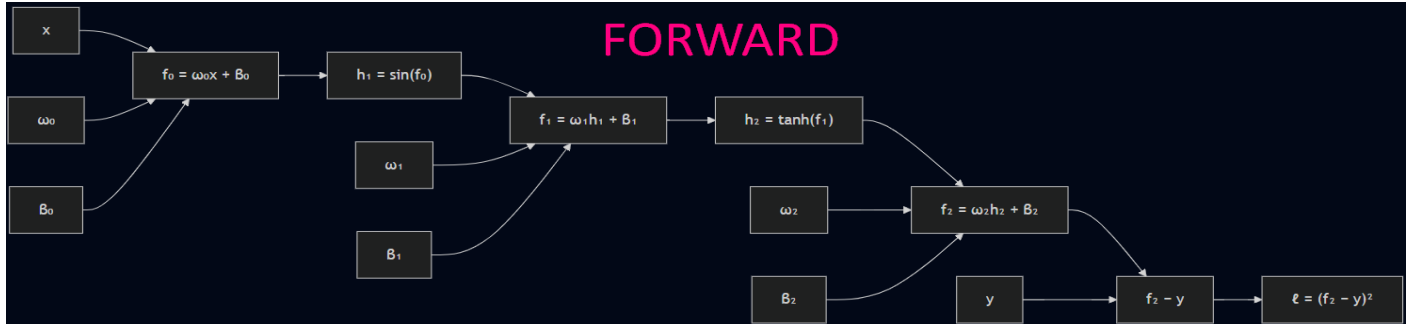
```
f1 --> h2["h2 = tanh(f1)"]
```

```
h2 --> f2["f2 = ω2h2 + β2"]
w2["ω2"] --> f2
b2["β2"] --> f2
```

```
f2 --> sub["f2 - y"]
```

```
y["y"] --> sub
```

```
sub --> loss["ℓ = (f2 - y)²"]
```



Grafo Backward

flowchart RL

```
loss["ℓ"] --> df2["∂ℓ/∂f2 = 2(f2 - y)"]
```

```
df2 --> dw2["∂ℓ/∂ω2 = 2(f2 - y)h2"]
```

```
df2 --> db2["∂ℓ/∂β2 = 2(f2 - y)"]
```

```
df2 --> dh2["∂ℓ/∂h2 = 2(f2 - y)ω2"]
```

```
dh2 --> df1["∂ℓ/∂f1 = 2(f2 - y)ω2(1 - h2²)"]
```

```
df1 --> dw1["∂ℓ/∂ω1 = 2(f2 - y)ω2(1 - h2²)h1"]
```

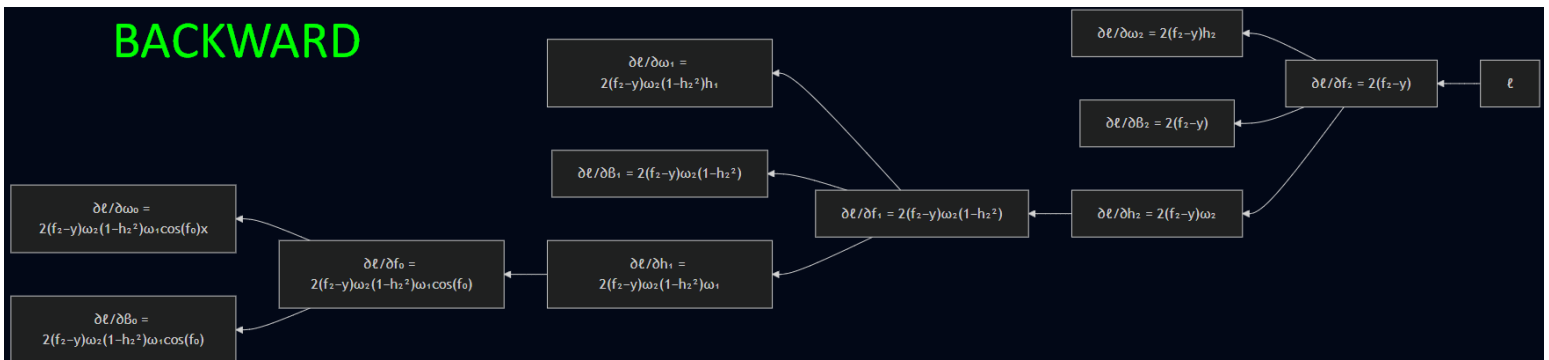
```
df1 --> db1["∂ℓ/∂β1 = 2(f2 - y)ω2(1 - h2²)"]
```

```
df1 --> dh1["∂ℓ/∂h1 = 2(f2 - y)ω2(1 - h2²)ω1"]
```

```
dh1 --> df0["∂ℓ/∂f0 = 2(f2 - y)ω2(1 - h2²)ω1cos(f0)"]
```

```
df0 --> dw0["∂ℓ/∂ω0 = 2(f2 - y)ω2(1 - h2²)ω1cos(f0)x"]
```

```
df0 --> db0["∂ℓ/∂β0 = 2(f2 - y)ω2(1 - h2²)ω1cos(f0)"]
```





Exercise 5: Two Hidden Neurons, One Input

Model:

$$f(x, \theta) = \omega_4 \tanh(\omega_0 x + \beta_0) + \omega_5 \sin(\omega_1 x + \beta_1) + \beta_2,$$

$$\theta = \{\omega_0, \beta_0, \omega_1, \beta_1, \omega_4, \omega_5, \beta_2\}.$$

Important: This model has two branches. During backpropagation, each branch must be handled separately.

1.- Forward Pass:

Rama 1: $f_0 = \omega_0 x + \beta_0$
 $h_1 = \tanh(f_0)$
 $a_1 = \omega_4 h_1$

Rama 2: $f_1 = \omega_1 x + \beta_1$
 $h_2 = \sin(f_1)$
 $a_2 = \omega_5 h_2$

Salida: $f_2 = a_1 + a_2 + \beta_2$
 $\ell = (f_2 - y)^2 \therefore \ell = (a_1 + a_2 + \beta_2 - y)^2$

2.- Loss:

$$\ell = (\omega_4 \tanh(\omega_0 x + \beta_0) + \omega_5 \sin(\omega_1 x + \beta_1) + \beta_2 - y)^2$$

3.- Grafo "Forward"

4.- Backward Pass: $\frac{\partial \ell}{\partial f_2} = 2(f_2 - y)$ definimos:

$$\delta = 2(f_2 - y) \text{ entonces: } \frac{\partial \ell}{\partial f_2} = \delta$$

como:

$$f_2 = a_1 + a_2 + \beta_2$$

$$\frac{\partial \ell}{\partial a_1} = \delta$$

$$\frac{\partial \ell}{\partial a_2} = \delta$$

$$\frac{\partial \ell}{\partial \beta_2} = \delta$$



Rama 1: Tanh

$$a_1 = w_4 h_1$$

$$h_1 = \tanh(f_0)$$

$$\frac{\partial L}{\partial w_4} = \delta h_1$$

$$\text{como } \frac{\partial h_1}{\partial f_0} = 1 - h_1^2$$

$$\frac{\partial L}{\partial h_1} = \delta w_4$$

$$\frac{\partial L}{\partial f_0} = \delta w_4 (1 - h_1^2)$$

como:

$$f_0 = w_0 x + \beta_0$$

$$\frac{\partial L}{\partial w_0} = \delta w_4 (1 - h_1^2) x$$

$$\frac{\partial L}{\partial \beta_0} = \delta w_4 (1 - h_1^2)$$

Rama 2: Sin

$$a_2 = w_5 h_2$$

$$\frac{\partial L}{\partial w_5} = \delta h_2 \text{ y } \frac{\partial L}{\partial h_2} = \delta w_5$$

como:

$$h_2 = \sin(f_1)$$

$$\frac{\partial h_2}{\partial f_1} = \cos(f_1)$$

$$\frac{\partial L}{\partial f_1} = \delta w_5 \cos(f_1)$$

como:

$$f_1 = w_1 x + \beta_1$$

$$\frac{\partial L}{\partial w_1} = \delta w_5 \cos(f_1) x \text{ y } \frac{\partial L}{\partial \beta_1} = \delta w_5 \cos(f_1)$$

5.- Gráfico "Backward" → Se envió como imagen



6.- Gradientes Finales

$$J = 2(f_2 - y)$$

$$f_0 = w_0 x + \beta_0$$

$$h_1 = \tanh(f_0)$$

$$f_1 = w_1 x + \beta_1$$

$$h_2 = \sin(f_1)$$

$$f_2 = w_4 h_1 + w_5 h_2 + \beta_2$$

Los gradientes son:

$$\frac{\partial L}{\partial w_0} = \delta w_4 (1 - h_1^2) x$$

$$\frac{\partial L}{\partial \beta_0} = \delta w_4 (1 - h_1^2)$$

$$\frac{\partial L}{\partial w_1} = \delta w_5 \cos(f_1) x$$

$$\frac{\partial L}{\partial \beta_1} = \delta w_5 \cos(f_1)$$

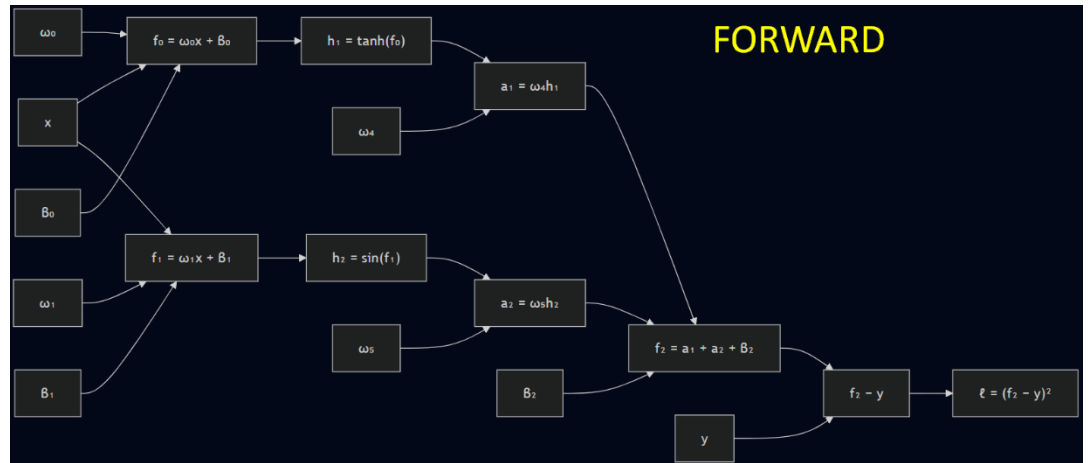
$$\frac{\partial L}{\partial w_4} = \delta h_1 \quad y \quad \frac{\partial L}{\partial w_5} = \delta h_2 \quad y \quad \frac{\partial L}{\partial \beta_2} = \delta$$



Grafo Forward

```

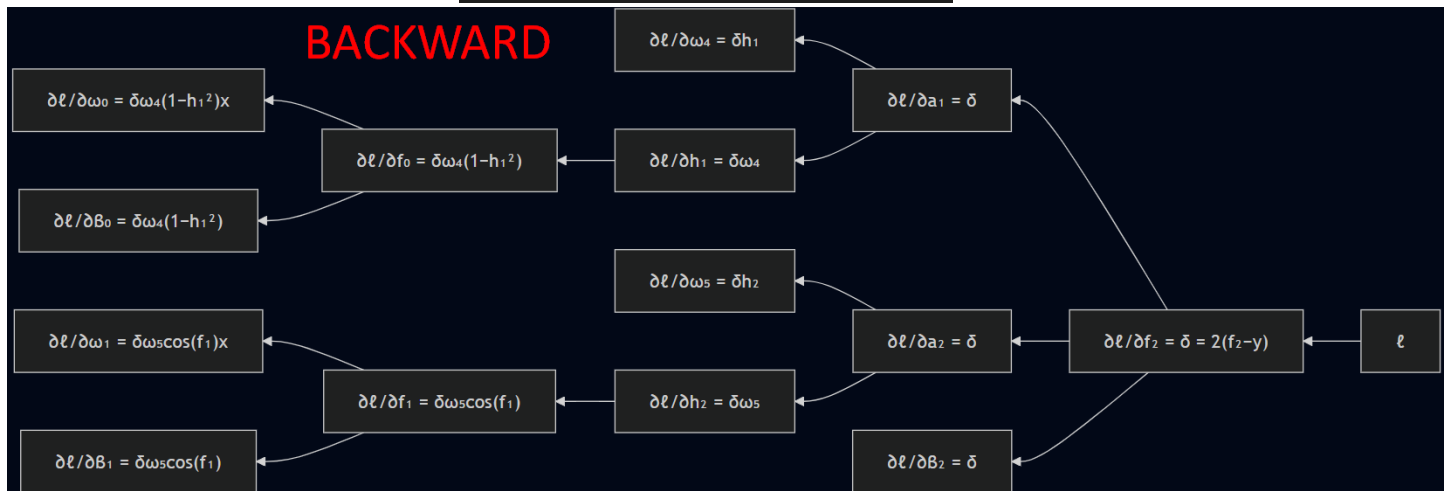
flowchart LR
    x["x"] --> f0["f0 = ω0x + β0"]
    w0["ω0"] --> f0
    b0["β0"] --> f0
    f0 --> h1["h1 = tanh(f0)"]
    h1 --> a1["a1 = ω4h1"]
    w4["ω4"] --> a1
    a1 --> f2["f2 = a1 + a2 + β2"]
    a2 --> f2
    w5["ω5"] --> a2
    B2["β2"] --> f2
    f2 --> y["y"]
    y --> loss["ℓ = (f2 - y)²"]
    
```



Grafo Backward

```

flowchart RL
    loss["ℓ"] --> df2["∂ℓ/∂f2 = δ = 2(f2 - y)"]
    df2 --> da1["∂ℓ/∂a1 = δ"]
    df2 --> da2["∂ℓ/∂a2 = δ"]
    df2 --> db2["∂ℓ/∂β2 = δ"]
    da1 --> dh1["∂ℓ/∂h1 = δω4"]
    da1 --> dw4["∂ℓ/∂ω4 = δh1"]
    dh1 --> df0["∂ℓ/∂f0 = δω4(1-h1²)"]
    df0 --> dw0["∂ℓ/∂ω0 = δω4(1-h1²)x"]
    df0 --> db0["∂ℓ/∂β0 = δω4(1-h1²)"]
    dh2["∂ℓ/∂h2 = δω5"] --> df1["∂ℓ/∂f1 = δω5cos(f1)"]
    df1 --> dw1["∂ℓ/∂ω1 = δω5cos(f1)x"]
    df1 --> db1["∂ℓ/∂β1 = δω5cos(f1)"]
    
```





Exercise 6: MLP with Two Inputs

Input:

$$x = (x_1, x_2).$$

Model:

$$f(x, \theta) = \omega_3 \tanh(\omega_0 x_1 + \omega_1 x_2 + \beta_0) + \beta_1,$$

$$\theta = \{\omega_0, \omega_1, \beta_0, \omega_3, \beta_1\}.$$

1.- Forward Pass:

$$f_0 = \omega_0 x_1 + \omega_1 x_2 + \beta_0$$

$$h_1 = \tanh(f_0)$$

$$f_1 = \omega_3 h_1 + \beta_1$$

$$\mathcal{L} = (f_1 - y)^2 = ((\omega_3 h_1 + \beta_1) - y)^2$$

2.- Loss

$$\mathcal{L} = (\omega_3 \tanh(\omega_0 x_1 + \omega_1 x_2 + \beta_0) + \beta_1 - y)^2$$

3.- Grafo "Forward" → Se envió como imagen

4.- Backward Pass →

$$\frac{\partial \mathcal{L}}{\partial f_1} = 2(f_1 - y)$$

$$\therefore \frac{\partial \mathcal{L}}{\partial f_1} = \delta$$

$$\delta = 2(f_1 - y)$$

$$f_1 = \omega_3 h_1 + \beta_1; \quad \frac{\partial \mathcal{L}}{\partial \omega_3} = \delta h_1; \quad \frac{\partial \mathcal{L}}{\partial \beta_1} = \delta; \quad \frac{\partial \mathcal{L}}{\partial h_1} = \delta \omega_3$$

$$h_1 = \tanh(f_0); \quad \frac{\partial h_1}{\partial f_0} = 1 - h_1^2; \quad \frac{\partial \mathcal{L}}{\partial f_0} = \delta \omega_3 (1 - h_1^2)$$

$$f_0 = \omega_0 x_1 + \omega_1 x_2 + \beta_0; \quad \frac{\partial \mathcal{L}}{\partial \omega_0} = \delta \omega_3 (1 - h_1^2) x_1; \quad \frac{\partial \mathcal{L}}{\partial \omega_1} = \delta \omega_3 (1 - h_1^2) x_2;$$

$$\frac{\partial \mathcal{L}}{\partial \beta_0} = \delta \omega_3 (1 - h_1^2)$$

5.- Grafo "Backward" → Se envió como imagen



6.- Gradientes Finales: $J = 2(f_1 - y)$

$$f_0 = w_0 x_1 + w_1 x_2 + \beta_0$$

$$h_1 = \tanh(f_0)$$

$$f_1 = w_3 h_1 + \beta_1 \quad \text{entonces:}$$

$$\frac{\partial L}{\partial w_0} = \delta w_3 (1 - h_1^2) x_1; \quad \frac{\partial L}{\partial w_1} = \delta w_3 (1 - h_1^2) x_2$$

$$\frac{\partial L}{\partial \beta_0} = \delta w_3 (1 - h_1^2); \quad \frac{\partial L}{\partial w_3} = \delta h_1$$

$$\frac{\partial L}{\partial \beta_1} = \delta$$



Grafo Forward

flowchart LR

```
x1["x1"] --> f0["f0 = ω0x1 + ω1x2 + β0"]
x2["x2"] --> f0
w0["ω0"] --> f0
w1["ω1"] --> f0
b0["β0"] --> f0
```

```
f0 --> h1["h1 = tanh(f0)"]
```

```
h1 --> f1["f1 = ω3h1 + β1"]
```

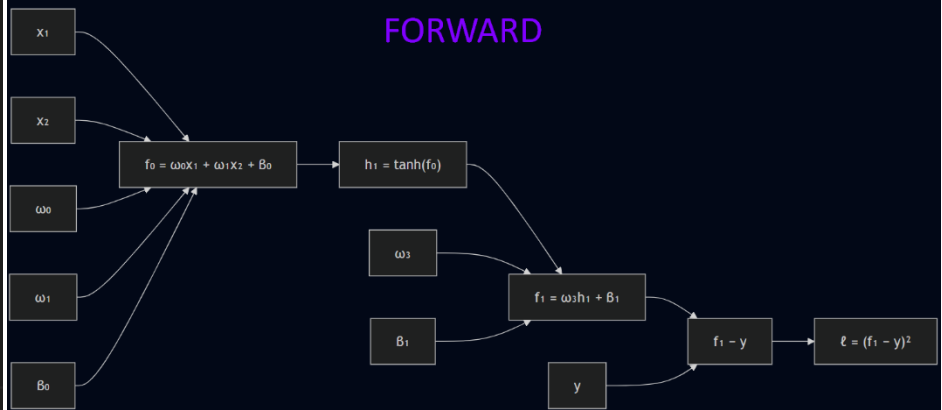
```
w3["ω3"] --> f1
```

```
b1["β1"] --> f1
```

```
f1 --> sub["f1 - y"]
```

```
y["y"] --> sub
```

```
sub --> loss["ℓ = (f1 - y)2"]
```



Grafo Backward

flowchart RL

```
loss["ℓ"] --> df1["∂ℓ/∂f1 = δ = 2(f1 - y)"]
```

```
df1 --> dw3["∂ℓ/∂ω3 = δh1"]
```

```
df1 --> db1["∂ℓ/∂β1 = δ"]
```

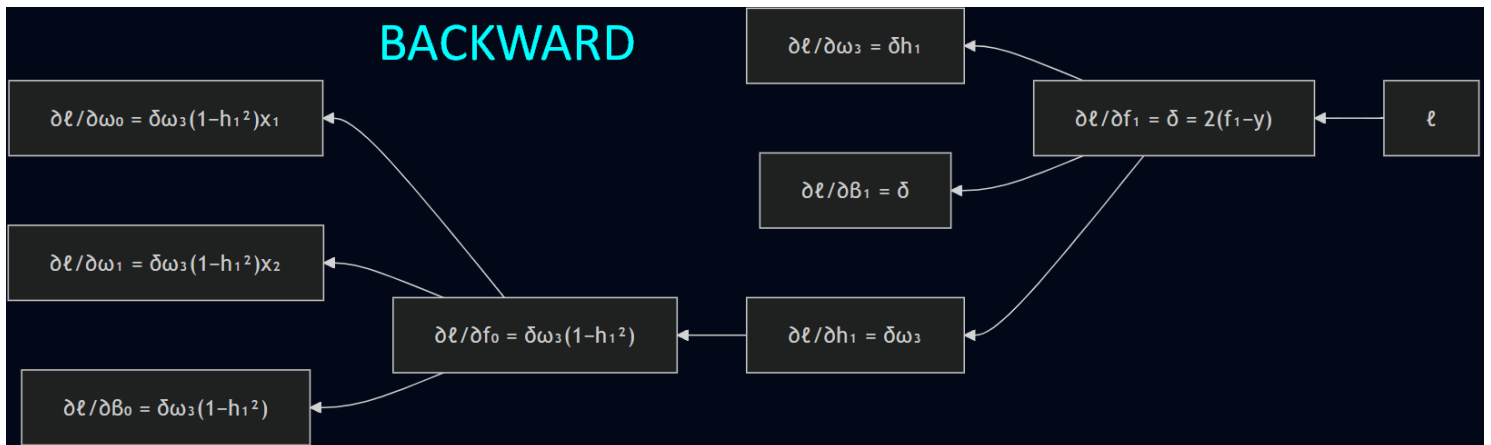
```
df1 --> dh1["∂ℓ/∂h1 = δω3"]
```

```
dh1 --> df0["∂ℓ/∂f0 = δω3(1 - h12)"]
```

```
df0 --> dw0["∂ℓ/∂ω0 = δω3(1 - h12)x1"]
```

```
df0 --> dw1["∂ℓ/∂ω1 = δω3(1 - h12)x2"]
```

```
df0 --> db0["∂ℓ/∂β0 = δω3(1 - h12)"]
```





Exercise 7: MLP 2-2-1

Input:

$$x = (x_1, x_2).$$

Model:

$$f(x, \theta) = \omega_5 \tanh(\omega_0 x_1 + \omega_1 x_2 + \beta_0) + \omega_6 \text{ReLU}(\omega_2 x_1 + \omega_3 x_2 + \beta_1) + \beta_2$$

$$\theta = \{\omega_0, \omega_1, \beta_0, \omega_2, \omega_3, \beta_1, \omega_5, \omega_6, \beta_2\}.$$

1.- Forward Pass

#1 Neurona Oculta: $f_0 = \omega_0 x_1 + \omega_1 x_2 + \beta_0$; $h_1 = \tanh(f_0)$

#2 Neurona Oculta: $f_1 = \omega_2 x_1 + \omega_3 x_2 + \beta_1$; $h_2 = \text{ReLU}(f_1) = \max(0, f_1)$

Salida: $f_2 = \omega_5 h_1 + \omega_6 h_2 + \beta_2$; $\mathcal{L} = (f_2 - y)^2$

2.- Loss: $\mathcal{L} = (\omega_5 \tanh(\omega_0 x_1 + \omega_1 x_2 + \beta_0) + \omega_6 \text{ReLU}(\omega_2 x_1 + \omega_3 x_2 + \beta_1) + \beta_2 - y)^2$

3.- Gráfico "Forward" → Se envía como imagen

4.- Backward Pass → $\delta = \frac{\partial \mathcal{L}}{\partial f_2} = 2(f_2 - y)$ como:

$f_2 = \omega_5 h_1 + \omega_6 h_2 + \beta_2$ entonces:

$$\frac{\partial \mathcal{L}}{\partial \omega_5} = \delta h_2; \quad \frac{\partial \mathcal{L}}{\partial \omega_6} = \delta h_2; \quad \frac{\partial \mathcal{L}}{\partial \beta_2} = \delta; \quad \frac{\partial \mathcal{L}}{\partial h_1} = \delta \omega_5; \quad \frac{\partial \mathcal{L}}{\partial h_2} = \delta \omega_6$$

Nota 1: $\tanh$

$h_1 = \tanh(f_0)$; $\frac{\partial h_1}{\partial f_0} = 1 - h_1^2$; $\frac{\partial \mathcal{L}}{\partial f_0} = \delta \omega_5 (1 - h_1^2)$ como:

$f_0 = \omega_0 x_1 + \omega_1 x_2 + \beta_0$; $\frac{\partial \mathcal{L}}{\partial \omega_0} = \delta \omega_5 (1 - h_1^2) x_1$; $\frac{\partial \mathcal{L}}{\partial \omega_1} = \delta \omega_5 (1 - h_1^2) x_2$

$$\frac{\partial \mathcal{L}}{\partial \beta_0} = \delta \omega_5 (1 - h_1^2)$$

Nota 2: ReLU

$h_2 = \text{ReLU}(f_1)$

La derivada ReLU es; $\frac{\partial h_2}{\partial f_1} = \begin{cases} 1, & f_1 > 0 \\ 0, & f_1 \leq 0 \end{cases}$

se transcribe como: $I(f_1 > 0)$ entonces $\frac{\partial \mathcal{L}}{\partial f_1} = \delta \omega_6 I(f_1 > 0)$



Como $f_1 = w_2 x_1 + w_3 x_2 + b_1$

$$\frac{\partial L}{\partial w_2} = \delta w_6 I(f_1 > 0) x_1$$

$$\frac{\partial L}{\partial w_3} = \delta w_6 I(f_1 > 0) x_2$$

$$\frac{\partial L}{\partial b_1} = \delta w_6 \pm (f_1 > 0)$$

5.- Grafo "Backward" → Se envió como imagen

6.- Gradientes Finales:

$$\frac{\partial L}{\partial w_0} = 2(f_2 - y) w_5 (1 - h_1^2) x_1$$

$$\frac{\partial L}{\partial w_1} = 2(f_2 - y) w_5 (1 - h_1^2) x_2$$

$$\frac{\partial L}{\partial b_0} = 2(f_2 - y) w_5 (1 - h_1^2)$$

$$\frac{\partial L}{\partial w_2} = 2(f_2 - y) w_6 I(f_1 > 0) x_1; \quad \frac{\partial L}{\partial w_3} = 2(f_2 - y) w_6 I(f_1 > 0) x_2$$

$$\frac{\partial L}{\partial b_1} = 2(f_2 - y) w_6 I(f_1 > 0); \quad \frac{\partial L}{\partial w_5} = 2(f_2 - y) h_1; \quad \frac{\partial L}{\partial w_6} = 2(f_2 - y) h_2$$

$$\frac{\partial L}{\partial b_2} = 2(f_2 - y)$$

donde:

$$f_0 = w_0 x_1 + w_1 x_2 + b_0$$

$$h_1 = \tanh(f_0)$$

$$h_2 = \text{ReLU}(f_1)$$

$$f_1 = w_2 x_1 + w_3 x_2 + b_1$$

$$f_2 = w_5 h_1 + w_6 h_2 + b_2$$



Grafo Forward

```

flowchart LR
    x1["x1"] --> f0["f0 = ω0x1 + ω1x2 + β0"]
    x2["x2"] --> f0
    w0["ω0"] --> f0
    w1["ω1"] --> f0
    b0["β0"] --> f0

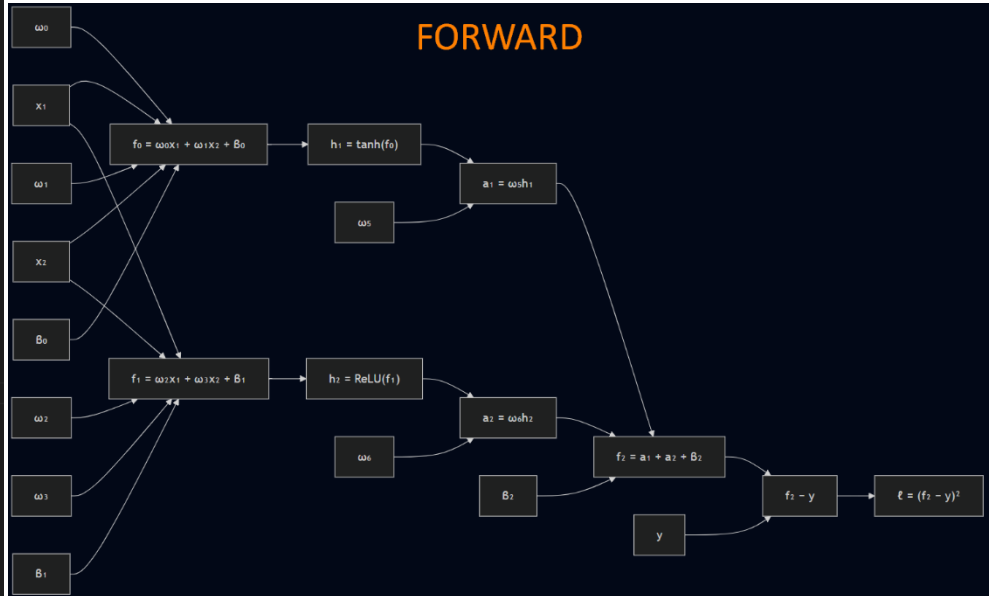
    f0 --> h1["h1 = tanh(f0)"]
    h1 --> a1["a1 = ω5h1"]
    w5["ω5"] --> a1

    x1 --> f1["f1 = ω2x1 + ω3x2 + β1"]
    x2 --> f1
    w2["ω2"] --> f1
    w3["ω3"] --> f1
    b1["β1"] --> f1

    f1 --> h2["h2 = ReLU(f1)"]
    h2 --> a2["a2 = ω6h2"]
    w6["ω6"] --> a2

    a1 --> f2["f2 = a1 + a2 + β2"]
    a2 --> f2
    b2["β2"] --> f2

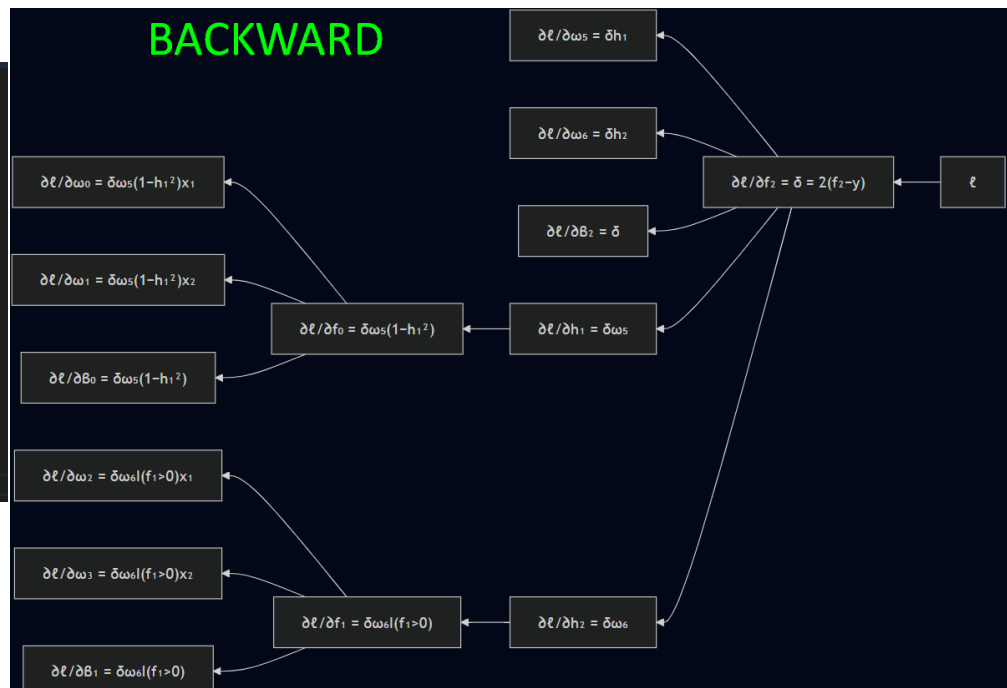
    f2 --> sub["sub f2 - y"]
    y["y"] --> sub
    sub --> loss["ℓ = (f2 - y)²"]
  
```



Grafo Backward

```

flowchart RL
    loss["ℓ"] --> df2["∂ℓ/∂f2 = δ = 2(f2 - y)"]
    df2 --> dh1["∂ℓ/∂h1 = δω5"]
    df2 --> dh2["∂ℓ/∂h2 = δω6"]
    df2 --> df0["∂ℓ/∂f0 = δω5(1 - h1²)"]
    df2 --> df1["∂ℓ/∂f1 = δω6I(f1 > 0)"]
    df0 --> dw0["∂ℓ/∂ω0 = δω5(1 - h1²)x1"]
    df0 --> dw1["∂ℓ/∂ω1 = δω5(1 - h1²)x2"]
    df0 --> db0["∂ℓ/∂β0 = δω5(1 - h1²)"]
    df1 --> dw2["∂ℓ/∂ω2 = δω6I(f1 > 0)x1"]
    df1 --> dw3["∂ℓ/∂ω3 = δω6I(f1 > 0)x2"]
    df1 --> db1["∂ℓ/∂β1 = δω6I(f1 > 0)"]
  
```





Exercise 8: Branching Model with Shared Pre-Activation

Model:

$$f(x, \theta) = \omega_3 \sin(\omega_0 x + \beta_0) + \omega_4 \cos(\omega_0 x + \beta_0) + \beta_1,$$

$$\theta = \{\omega_0, \beta_0, \omega_3, \omega_4, \beta_1\}.$$

Important: The same node $\omega_0 x + \beta_0$ is used by both $\sin(\cdot)$ and $\cos(\cdot)$. During the backward pass, gradients arriving at this shared node must be added. Explicitly show where gradient accumulation occurs.

1.- Forward Pass

"El nodo": $f_0 = \omega_0 x + \beta_0$
se comparte entre
las 2 ramas:

$$h_1 = \sin(f_0); \quad h_2 = \cos(f_0); \quad a_1 = \omega_3 h_1; \quad a_2 = \omega_4 h_2;$$

$$f_1 = a_1 + a_2 + \beta_1; \quad \mathcal{L} = (f_1 - y)^2$$

2.- Loss: $\mathcal{L} = (\omega_3 \sin(\omega_0 x + \beta_0) + \omega_4 \cos(\omega_0 x + \beta_0) + \beta_1 - y)^2$

3.- Grafo "Forward" → Se envió como imagen

4.- Backward Pass → $\delta = \frac{\partial \mathcal{L}}{\partial f_1} = 2(f_1 - y)$
 $f_1 = a_1 + a_2 + \beta_1$

$$\frac{\partial \mathcal{L}}{\partial a_1} = \delta$$

$$\frac{\partial \mathcal{L}}{\partial a_2} = \delta$$

$$\frac{\partial \mathcal{L}}{\partial \beta_1} = \delta$$

Rama Seno: $a_1 = \omega_3 h_1; \quad \frac{\partial \mathcal{L}}{\partial \omega_3} = \delta h_1; \quad \frac{\partial \mathcal{L}}{\partial h_1} = \delta \omega_3$

$$h_1 = \sin(f_0); \quad \frac{\partial h_1}{\partial f_0} = \cos(f_0)$$

entonces la contribución de la rama seno hacia f_0 es:

$$\left(\frac{\partial \mathcal{L}}{\partial f_0} \right)_{\sin} = \delta \omega_3 \cos(f_0)$$

Rama Coseno: $a_2 = \omega_4 h_2; \quad \frac{\partial \mathcal{L}}{\partial \omega_4} = \delta h_2; \quad \frac{\partial \mathcal{L}}{\partial h_2} = \delta \omega_4$

$$h_2 = \cos(f_0) \quad \text{y} \quad \frac{\partial h_2}{\partial f_0} = -\sin(f_0)$$

entonces la contribución de rama coseno hacia f_0 es:

$$\left(\frac{\partial \mathcal{L}}{\partial f_0} \right)_{\cos} = -\delta \omega_4 \sin(f_0)$$



- Existe una acumulación "gradiente" en el nodo, como f_0 alimenta h_1 y h_2 los gradientes se suman?

$$\frac{\partial L}{\partial f_0} = \left(\frac{\partial L}{\partial f_0} \right)_{\text{sen}} + \left(\frac{\partial L}{\partial f_0} \right)_{\text{cos}}$$

$$\frac{\partial L}{\partial f_0} = \delta w_3 \cos(f_0) - \delta w_4 \sin(f_0)$$

$$\frac{\partial L}{\partial f_0} = \delta (w_3 \cos(f_0) - w_4 \sin(f_0)) \rightarrow f_0 = w_0 x + \beta_0$$

$$\frac{\partial L}{\partial w_0} = \delta (w_3 \cos(f_0) - w_4 \sin(f_0)) x$$

$$\frac{\partial L}{\partial \beta_0} = \delta (w_3 \cos(f_0) - w_4 \sin(f_0))$$

5.- Grafo "Backward" $\rightarrow$ se envia como imagen

6.- Gradientes Finales:

$$\begin{aligned} \delta &= 2(f_1 - y) & h_1 &= \sin(f_0) \\ f_0 &= w_0 x + \beta_0 & h_2 &= \cos(f_0) \\ f_1 &= w_3 h_1 + w_4 h_2 + \beta_1 \end{aligned}$$

$$\frac{\partial L}{\partial w_0} = 2(f_1 - y)(w_3 \cos(f_0) - w_4 \sin(f_0)) x$$

$$\frac{\partial L}{\partial \beta_0} = 2(f_1 - y)(w_3 \cos(f_0) - w_4 \sin(f_0))$$

$$\frac{\partial L}{\partial w_3} = 2(f_1 - y) \sin(f_0)$$

$$\frac{\partial L}{\partial w_4} = 2(f_1 - y) \cos(f_0)$$

$$\frac{\partial L}{\partial \beta_1} = 2(f_1 - y)$$

donde $f_0 = w_0 x + \beta_0$
 $f_1 = w_3 \sin(f_0) + w_4 \cos(f_0)$



Grafo Forward

```

flowchart LR
    x["x"] --> f0["f0 = ω0x + β0"]
    w0["ω0"] --> f0
    b0["β0"] --> f0

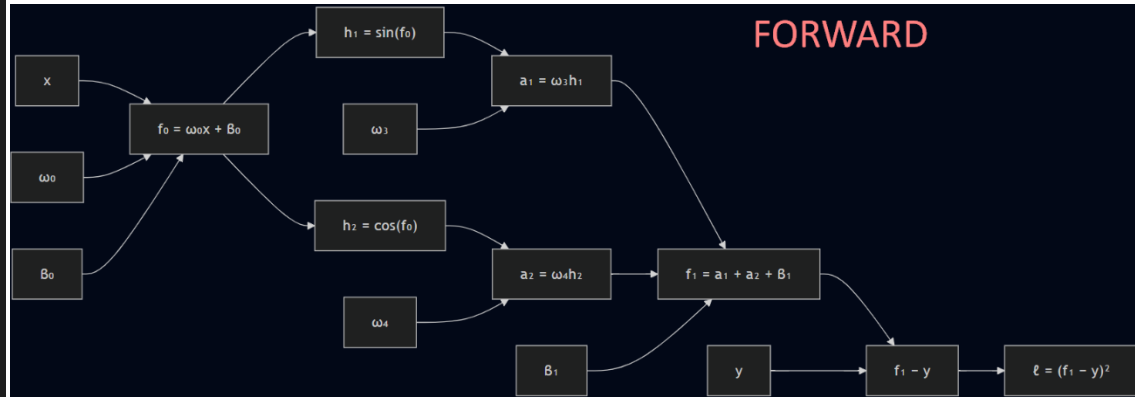
    f0 --> h1["h1 = sin(f0)"]
    f0 --> h2["h2 = cos(f0)"]

    h1 --> a1["a1 = ω3h1"]
    w3["ω3"] --> a1

    h2 --> a2["a2 = ω4h2"]
    w4["ω4"] --> a2

    a1 --> f1["f1 = a1 + a2 + β1"]
    a2 --> f1
    b1["β1"] --> f1

    f1 --> sub["f1 - y"]
    y["y"] --> sub
    sub --> loss["ℓ = (f1 - y)²"]
  
```



Grafo Backward

```

flowchart RL
    loss["ℓ"] --> df1["∂ℓ/∂f1 = δ = 2(f1 - y)"]
    df1 --> da1["∂ℓ/∂a1 = δ"]
    df1 --> da2["∂ℓ/∂a2 = δ"]
    df1 --> db1["∂ℓ/∂β1 = δ"]

    da1 --> dw3["∂ℓ/∂ω3 = δh1"]
    da1 --> dh1["∂ℓ/∂h1 = δω3"]
    dh1 --> contrib1["rama sin: δω3cos(f0)"]

    da2 --> dw4["∂ℓ/∂ω4 = δh2"]
    da2 --> dh2["∂ℓ/∂h2 = δω4"]
    dh2 --> contrib2["rama cos: δω4sin(f0)"]

    contrib1 --> acc["Acumulación en f0"]
    contrib2 --> acc

    acc --> df0["∂ℓ/∂f0 = δ(ω3cos(f0) - ω4sin(f0))"]
    df0 --> dw0["∂ℓ/∂ω0 = δ(ω3cos(f0) - ω4sin(f0))x"]
    df0 --> db0["∂ℓ/∂β0 = δ(ω3cos(f0) - ω4sin(f0))"]
  
```

BACKWARD

